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TAPS: Tailored Auditory Precision Stimulation

Transforming sound into a personalised digital medicine: sound therapy for tinnitus.

Summary

More than 740 million adults live with tinnitus, a sound heard without an external source, and over 120 million are severely affected. Tinnitus erodes sleep, concentration and work, and is linked to anxiety, depression and suicidal thoughts. There is no cure, and most current treatments help people manage their reaction to tinnitus, while the tinnitus itself remains.

Sound therapy is a highly accessible, non-invasive treatment option, and acts on the auditory pathways that generate tinnitus. A short sound can reduce tinnitus for seconds to minutes afterwards, an effect called residual inhibition. I propose to develop TAPS, a programme to test whether that brief reduction can be turned into lasting relief, tailored to each individual's unique tinnitus. In TAPS, each person's therapeutic stimulus will be chosen by testing which stimuli best reduce that person's tinnitus, embedded in sounds the person enjoys, and updated as a phone app records listening hours and changes in the tinnitus. TAPS will improve on current sound therapy in two ways: the sound is chosen and updated from each person's own measured response, and it sits inside listening the person already enjoys, so people keep listening. Early studies show the promise of each of these steps, which have not yet been joined into one comprehensive solution.

TAPS will run as a challenge: pioneer teams will compete on developing technologies for precision sound stimuli, delivery and engagement, while measurement, the trial protocol, validation and an open dataset are built by funded expert teams and run centrally. This shared core is a resource no single company could produce or fund. The programme will run in three stages, each with a milestone, over three years at a cost of €15 to 24 million. **If TAPS succeeds, millions of people with tinnitus could receive a treatment that reduces their tinnitus, on the phones and headphones they already own and use, backed by evidence that opens a new field of precision digital sound therapy. At minimum, the programme produces an open dataset and a phone-based way to test sound therapies for multiple conditions that others can build on.**

1. A common, costly and neglected condition

Tinnitus is a ringing, hissing or buzzing heard without an external source, due to disrupted processing in the brain's auditory pathways. Tinnitus usually follows damage to the ear, from noise, ageing or injury, with or without measurable hearing loss. When the ear sends fewer signals at some frequencies, the auditory brain turns up its own sensitivity, and the resulting spontaneous hyperactivity in the auditory pathways is the leading explanation for the phantom sound[1].

In Germany, the global prevalence rates[2] mean roughly 10 million adults live with tinnitus, about 1.6 million of them affected severely, costing Germany an estimated €22 to 35 billion a year. About half of this cost comes from sick leave: people with chronic tinnitus take about 26 sick days a year against a national average of 11[3]. In the US, tinnitus is the most common service-connected disability among veterans, about 3.3 million of whom are compensated for it[4]. These figures are likely underestimates: the German estimate comes from one centre at 2017 costs, and global burden statistics record tinnitus only as part of hearing loss[5].

2. Why current treatments fall short

Few current treatments act on the auditory pathways directly. Antidepressants can ease depression and anxiety, while the tinnitus itself continues[6]. Cognitive behavioural therapy (CBT) helps people manage their response to the sound, but the perceived sound remains. Devices that pair sound with electrical stimulation, such as Lenire, show promise[7,8], but need dedicated hardware and clinic fitting, at significant cost. Sound therapy, which delivers prescribed sound for long periods each day and acts on the auditory pathways directly, is the focus of TAPS. Reducing distress and reducing the tinnitus itself are different outcomes, and TAPS will measure both. The Tinnitus Functional Index (TFI), a validated

questionnaire on how tinnitus affects daily life[9], provides the primary measure of effects on sleep, concentration and mood. Tinnitus loudness, rated each day before listening, is the measure of the sensation itself.

3. Sound therapy: two problems, two fixes

TAPS will test whether the brief reduction in the tinnitus itself seen in residual inhibition[10] can be turned into lasting change through regular listening over weeks. In a multicentre, double-blind, placebo-controlled trial, 12 weeks of audiometrically filtered noise (noise shaped to each person's hearing test so that all frequencies are equally audible) reduced TFI scores by 21 points against 6 with placebo, and 88% of participants reached a clinically meaningful improvement against 33%[11]. A blinded online trial on participants' own devices found a reduction that persisted after listening stopped, with responses varying widely between participants[12]. TAPS will build on this promise and tailor the sound stimulation to each person's response (Table 1).

Table 1. Current sound therapy options: two problems, two fixes

	Current problem	TAPS solution	What is tested
More effective sound stimulation	Sound usually chosen once, from a hearing test or pitch match Rarely adapted or optimised afterwards[1].	1. Test many sounds 2. Choose those that best reduce the tinnitus 3. Re-test and update over time	Lasting benefit beyond the sounds that carry the stimulus and a control sound (Table 3, Appendix A1) Residual inhibition as a predictor of benefit[10,13]
Increase listening and engagement	Sound often unpleasant: people stop listening within weeks[14,15] Listening time rarely measured	1. Embed therapeutic stimuli inside sounds the person enjoys, using generative AI 2. Log listening and activity (wearables) 3. Keep people engaged with gamified check-ins	Do people listen more? Does more listening bring more benefit?[16] Do the same sounds without the stimulus help less? Wellbeing: TFI and wearables

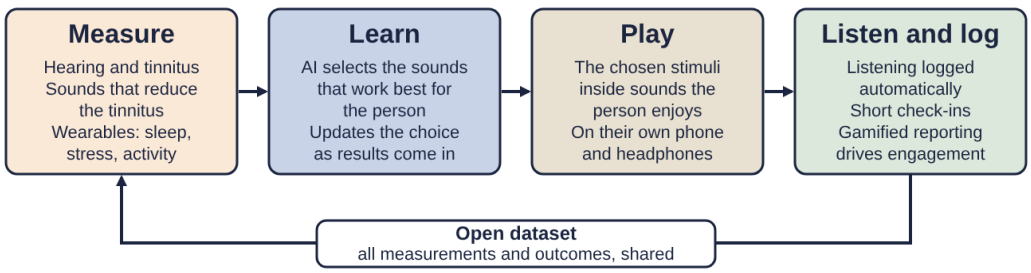
4. Why this is possible now

Three capabilities can now be combined into one learning system. Some smartphone hearing tests match clinic tests to within 10 decibels (dB)[17]. Tinnitus reduction can be measured within minutes, with a standard clinical test, which TAPS would move to a phone[18]. Generative AI can produce music to precise specifications[19], so a therapeutic stimulus can be built into sounds people enjoy, keeping them listening. AI runs through the whole loop: measuring, selecting, generating and adapting. Gamified check-ins keep people reporting their tinnitus and its effects on daily life via the TFI, and wearables add continuous data on sleep, activity and stress that does not depend on self-report. TAPS would run on the phones and consumer headphones most people already own, so the therapy could be widely accessible. Phone hearing tests depend on accurately set sound levels, so the shared core provides a per-device calibration tool (Appendix A4).

5. What TAPS will build, and how the work is organised

TAPS will be a learning loop that runs on a phone (Figure 1), with three objectives. (1) An individual auditory fingerprint: a phone app will measure each person's hearing and tinnitus, and test which sounds reduce that person's tinnitus (residual inhibition), with AI steering the search (Appendix A5). (2) Tailored sound in everyday listening: a therapeutic stimulus generated from those measurements and placed inside sounds the person chooses, with listening logged by the app. (3) Lasting relief, measured in an engaging way: a controlled comparison will test whether the tinnitus is reduced and stays reduced, and who benefits most. Daily-life impact will be measured by the TFI, a game-based version of it under test, and wearables.

Figure 1. The TAPS learning loop



As programme director I would run the shared core and, in Stage 1, select and fund expert teams to build that core to a set specification: the phone measurement module, the calibration tool, one trial protocol, validation and the open dataset. SPRIND pioneer teams will compete on three challenges (Table 2), all using the shared tools, and some Stage 1 expert teams may go on to compete as pioneer teams. Stages and milestones are in Table 3 (Appendix A1), the Gantt and budget in Appendices B and C.

Table 2. Three challenges for pioneer teams

Challenge	What teams will build	How success is measured
Sound stimuli	Sounds the person enjoys, carrying the selected stimulus Stimulus adapted as the person responds	Stimulus verified in the rendered audio and by a reduction test Greater TFI reduction vs the plain stimulus alone Enjoyment rating
Delivery	The therapy on any phone and ordinary consumer headphones, calibrated per model Uses the shared calibration tool; safe levels Integration in everyday listening	Sound level at the ear within tolerance across common device pairs Share of listening inside everyday sounds No safety-cap breaches
Engagement	Ways to keep people listening and reporting: Onboarding and habit design Gamified check-ins built on the TFI Wearables	Retention over time Days listened Check-ins completed Gamified versus standard TFI accuracy

Four products will be produced: (1) a validated phone measurement app, for patients, clinics and researchers; (2) AI sound generation technology, for any sound-therapy product; (3) controlled clinical evidence, for regulators and reimbursement; (4) an open dataset and benchmarks, for auditory digital medicine developers.

6. Why public money, and the route to patients

The pieces of TAPS sit in three industries: audiology, consumer audio and generative AI audio. Start-ups struggle to fund trials whose results competitors would use for free, and device makers have little incentive to pay for shared evidence. Public funding fills that gap and leaves an open dataset for all three industries. What the €15 million and €24 million budgets would each fund is set out in Appendix C. In Germany, the DiGA directory of reimbursable digital health applications (DiGA, Digitale Gesundheitsanwendung) already lists a CBT-based tinnitus app, Kalmeda[20], and the DiGA fast-track offers a route for a personalised sound therapy. The UK brings the National Institute for Health and Care Excellence (NICE) evidence standards for digital health[21], and key clinical centres in both countries are well placed to partner on trials.

7. Beyond tinnitus

Tinnitus is where TAPS starts because it is a problem of sound: sound can target the tinnitus, and a person can tell within minutes whether the tinnitus is reduced. Hearing care is the nearest extension: hearing aids and other wearable audio devices already offer sound therapy. Hyperacusis, reduced tolerance of everyday sounds, is another possible extension[1]. The same loop (measure, personalise, deliver and adapt) could extend to other conditions where precisely specified sounds are under study: dementia[22], attention problems in young people[23] and gait in Parkinson's disease[24]. What TAPS would leave behind, a validated way to measure a person's response to sound on their own phone, a trial protocol and controlled evidence for sound as a treatment, and an open dataset, is the groundwork for digital sound medicine more widely.

8. Why me

I am an auditory neuroscientist with a PhD in computational neuroscience from Oxford. I co-founded NeuroFilt Audio, an audio machine-learning start-up, and at Faculty AI I lead the delivery of applied AI systems from prototype to production. This problem is also personal: I have tinnitus myself, and several people close to me do too. That is not why I think TAPS will work, but it is why I will not stop trying.

Appendix

Acknowledgements.
I am grateful for the input from the advisers whose expertise has strengthened, and will continue to strengthen, this programme and its products of benefit to society:

Jane Roskams, neuroscientist, University of British Columbia and Science Impact Studio, mentor on the Big if True Science Accelerator (BiTS) (programme design and positioning)
Martin Pienkowski, sound therapy and auditory neuroscience researcher, Osborne Audiology, Drexel University (sound therapy evidence and trial design)
James A. Henry, co-author of the Tinnitus Functional Index, formerly US Department of Veterans Affairs National Center for Rehabilitative Auditory Research (tinnitus measurement and the residual inhibition protocol)
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Stephen Friend, co-founder of Sage Bionetworks and 4YouandMe, Visiting Professor of Connected Medicine, University of Oxford (baseline design, wearables and physiological readouts)
Andrew Welchman, Professor of Neuroinnovation, University of Birmingham, formerly Head of Neuroscience and Mental Health at Wellcome (digital intervention studies and regulation)
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Appendix A. Programme design in detail

Table 3. Three stages, each with a milestone (stage budgets exclude the €1 to 2M open dataset line, Appendix C)

Stage, who and when	What happens	What is measured	Milestone
1. Measure and learn **Year 1, €5 to 8M** **About 100 adults, one site** **Funded expert teams build the shared core**	1. Phone measurement checked against clinic tests 2. Pre-testing baseline of daily tinnitus ratings 3. Tinnitus reduction tests (Appendix A5) find each person's best stimulus	Hearing via Audiogram Tinnitus pitch and loudness Day-to-day variation Which sounds reduce the tinnitus, by how much and for how long	End of year 1: Measurements repeat reliably Sounds that reduce the tinnitus found for most people
2. Adopt and engage **Year 2, €5 to 8M** **Pioneer teams compete; new people join**	1. AI-generated sounds enter 2. Crossover (within subject control): 6 weeks: sounds with stimulus 6 weeks: same sounds without Random order 3. Comparison group: plain stimulus alone	Logged listening Daily loudness TFI at 0, 6 and 12 weeks Enjoyment Gamified versus standard TFI Wearables	Week 12: People keep listening Selected stimulus helps more than the carrier sound alone Best sounds progress to Stage 3
3. Optimise sounds, listen and log **Year 3, €4 to 6M** **New participants, one or more sites**	1. Best personalised stimulus versus a control stimulus, both embedded in the same enjoyable sounds 2. 12 weeks plus 3 months follow-up 3. Fixed-dose and self-directed groups	TFI (primary) Daily loudness Listening logs Wearables 3-month follow-up	Result: Who benefits, by how much How much listening is enough

A1. Stages and milestones. In Stage 1, each person rates their tinnitus daily during a pre-testing baseline, before any therapeutic sound, so that a treatment signal can later be separated from natural day-to-day variation. Stage 2 uses a within-person crossover: each person is their own control. For 6 weeks they listen to sounds they enjoy with their selected stimulus embedded, and for 6 weeks to the same sounds without the stimulus, in an order set at random. The difference between the two periods shows what the stimulus adds, independent of differences between people. A parallel group listens to the plain stimulus alone, for the comparison outlined in Appendix A2. In Stage 3, groups with a fixed amount of listening and groups that choose their own listening time show how much listening is enough. Because people respond to different degrees, results are reported across the full range of responses. Each milestone sets the plan for the next stage: continue as designed, or adjust the plan (for Milestone 2, the fallback in A2). The data is published either way.

A2. Hypotheses and decision rules. The central hypothesis is that personalised, selected and adaptable sound brings more lasting benefit than a control sound: Stage 2 tests whether the selected stimulus adds benefit over the sounds alone, and Stage 3 tests the selected sound against a control sound. The second hypothesis is that the stimulus inside sounds the person enjoys brings more benefit than the plain stimulus because people listen more, even if the effect per hour of listening is no greater; Stage 2 tests that hypothesis against a group listening to the plain stimulus alone. If the selected stimulus does not add benefit at Milestone 2, sounds would be chosen from each person's tinnitus pitch and hearing profile instead. Whether people with stronger residual inhibition respond better overall would be tested as a pre-registered secondary analysis on the pooled data.

A3. Participants and safeguards against bias. Participants are adults with chronic tinnitus and hearing loss up to about 50 dB hearing level (HL) at the relevant frequencies, grouped by hearing loss at the tinnitus frequency and by residual inhibition. People with no measurable residual inhibition stay in the study: their sound is set from the tinnitus pitch match and hearing profile, as in current practice, and their results count towards the secondary analysis in A2. Early stages test where the therapeutic sound is most likely to be audible and effective; people with greater hearing loss may also benefit and could be included in later work. Enjoyable sounds with and without an embedded stimulus are hard to blind fully, and the TFI is a self-report, so the design would add four safeguards: the within-person crossover with matched sounds, blinded outcome assessment, an expectation measure at baseline, and loudness rated each day before listening, so that short-term relief is not mistaken for lasting change.

A4. Sounds, listening level and calibration. Sound therapy, and the search phase of the reduction test, run within a range of levels the person finds comfortable, while the first clinical test (Appendix A5) uses the published level; earlier studies indicate that the exact level matters little provided the sound is audible and partly masks the tinnitus[1]. Phone hearing tests depend on calibrated output, and consumer headphones vary most between 3 and 8 kilohertz (kHz), so the shared core provides a calibration profile for each supported headphone model, with early stages run on one or two models. TAPS would seek operating-system or manufacturer cooperation for device-level delivery.

A5. The reduction test and the search for each person's best stimulus. The reduction test is the core measurement of Stage 1: a short sound plays, and the person reports whether the tinnitus is reduced afterwards, by how much and for how long (residual inhibition). The test follows the standard clinical procedure[18]: wide-band noise is raised to the lowest level that covers the tinnitus, raised a further 10 dB if the person tolerates the increase, and played for 60 seconds. After the noise stops, the person rates the tinnitus every 10 to 15 seconds, as a percentage change from usual, until the tinnitus returns to its usual level. A shorter pulsed version[25] is also tried as the form best suited to a phone.

The reduction depends on the sound: on the frequency region of the noise relative to the tinnitus and hearing loss[26], and on level, duration and modulation, with large differences between people[27]. Stage 1 therefore describes each test sound by a few settings (centre frequency, bandwidth, level, duration, modulation) and searches those settings for each person. Each test gives one score, the reduction summed over the time until the tinnitus returns. A statistical model estimates, from the person's responses so far, the expected reduction for any settings and the uncertainty of that estimate, and each next test sound is chosen where the model expects the largest reduction or has the most to learn (Bayesian optimisation), the method already used to estimate hearing tests from far fewer tones[28]. Tests are repeated on separate days to check each person's result for day-to-day stability. Whether the stimulus selected by this test brings more lasting benefit than a control stimulus is the central hypothesis, tested in Stages 2 and 3 (Appendix A2).

Appendix B. Timeline in months: three stages with overlaps

Stage	1 to 6	7 to 12	13 to 18	19 to 24	25 to 30	31 to 36
Stage 1. Measure and learn		Milestone 1				
Stage 2. Adopt and engage		from month 9	Milestone 2			
Stage 3. Optimise sounds, listen and log				from month 22		follow-up
Open dataset and benchmark						

Expert teams for Stage 1 would be selected in months 1 to 3. Team selection and sound development for Stage 2 would begin before Milestone 1; participant listening would follow Milestone 1. Milestone criteria are in Table 3 (Appendix A1).

Appendix C. Budget ranges

Line	When	Range	What the range covers
Stage 1. Measure and learn	Year 1	€5 to 8M	Awards to selected expert teams Phone measurement and calibration tools Clinic validation Baseline phase Reduction tests with about 100 people at one site
Stage 2. Adopt and engage	Year 2	€5 to 8M	Pioneer team awards 6 plus 6 week crossover Engagement and wearable data layer Plain-stimulus comparison group
Stage 3. Optimise sounds, listen and log	Year 3	€4 to 6M	Controlled comparison with follow-up Fixed-dose and self-directed groups
Open dataset and benchmark	All stages	€1 to 2M	Data infrastructure Curation and access Benchmark releases
Total	About 3 years	€15 to 24M	€15M: three teams, one Stage 3 site €24M: five to six teams, several sites, wearables at scale Expert teams funded from month 1

Appendix D. Earlier work that TAPS will build on

In every row, TAPS will also add logged listening, continued adaptation and an open dataset.

Approach	What it does	What TAPS will add
Bimodal neuromodulation (Lenire)[7,8]	Sound set from the audiogram (hearing test), paired with tongue stimulation Dedicated device fitted in clinic US authorisation 2023, Canada 2026	Sound chosen by reduction test (residual inhibition) The person's own phone and the sounds they enjoy
Pitch-matched and notched music (for example AudioNotch[29])	Notch or boost at the matched tinnitus pitch in the user's own music Set once from a pitch match	Selection from measured response, in place of a single pitch match A control sound
Tones embedded in chosen music[30]	Four tones around the tinnitus frequency inside self-chosen classical music Fixed for 12 weeks, 25 completers	Sound chosen by reduction test Any sound the person enjoys Controlled evidence at scale
Sound selected by residual inhibition[25]	Reduction test by hand in a clinic booth 20 people, hearing aids Sound set once	The test on a phone, repeated over time Sounds the person chooses A control sound
Digital polytherapeutic (combining counselling and sound therapy)[31]	Counselling plus sound shaped to audiogram, pitch and preference Adjusted by a clinician	Measured-response selection Automatic adaptation
CBT, support and hearing-aid apps (Kalmeda, DiGA-listed[20]; Oto; ReSound Relief)	Behavioural support with sound tools Soundscapes chosen by the user	Acts on the tinnitus itself Measured selection Evidence on dose

Appendix E. Open design questions

To be co-designed with SPRIND, clinical advisers and people with lived experience of tinnitus:

- The reduction-test search and re-test schedule, beyond the starting protocol in Appendix A5
- The length of the pre-testing baseline
- A no-sound period in the Stage 2 crossover (baseline stability, durability of gains)
- The hearing-loss inclusion range
- The form of the control sound
- The expectation measure
- Which wearable signals to collect
- Sample sizes and success thresholds at each milestone
- Listening schedules for the fixed-dose group

Appendix F. Main risks

The main scientific risk is that the personalised sound brings no lasting benefit over the controls even when people listen enough; the Stage 2 and Stage 3 comparisons are designed to detect that outcome, and the dataset remains valuable either way. The main practical risk is insufficient listening, which Stage 2 tests before Stage 3 and the Engagement challenge addresses directly. Sound levels are capped to meet World Health Organization (WHO) safe-listening guidance.

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